

Most Organisations Cannot Safely Use AI

Not Because of the Models, but Because Their Work Is Not Actually Defined

Abstract

The current discourse around AI adoption treats the problem as one of tooling: model capability, prompt engineering, integration patterns, and cost. This framing is incomplete. The primary constraint on the safe and effective use of AI in knowledge work is not the capability of the models, but the degree to which the underlying human work has been explicitly defined.

Most organisational work is not defined in a way that can be safely externalised. It is held together by tacit knowledge, informal negotiation, and continuous human repair. AI systems do not participate in this repair. When introduced into such environments, they do not immediately fail; they produce outputs that are locally coherent while propagating global inconsistency.

This is not a failure of AI. It is an exposure of pre-existing architectural weakness.

1. The Illusion of AI Adoption as a Tooling Problem

AI adoption is commonly framed as a question of capability:

- Which model should we use?
- How do we integrate it into our stack?
- How do we write better prompts?

- How do we measure ROI?

These are practical questions, but they assume that the work being performed is already well-formed. That assumption is rarely examined.

In most organisations, work is not defined through explicit, stable structures. It is defined through a combination of:

- role descriptions that conflate multiple responsibilities
- documents that encode outcomes but not intent
- meetings that resolve ambiguity without recording it
- individuals who “know how things actually work”

This system functions because humans compensate for its gaps. They infer intent, resolve contradictions, and repair misunderstandings as they arise.

The organisation appears to operate coherently. In reality, coherence is continuously reconstructed.

2. The Hidden Dependency: Human Repair

Human workflows depend on a property that is rarely named: the ability to repair ambiguity.

When a requirement is underspecified, a developer asks a question.

When two documents conflict, a decision is made in a meeting.

When behaviour diverges from expectation, someone recognises it and intervenes.

These repairs are:

- local (they occur in specific interactions)
- ephemeral (they are rarely recorded in full)
- social (they depend on shared context and judgement)

Over time, they accumulate into a functioning system. But the system itself is not explicit. It exists partly in artefacts and partly in the heads of the people maintaining it.

This is why two teams can follow the same documented process and produce different results. The documentation is not the system. The repair mechanisms are.

3. What AI Actually Does to This System

AI systems do not participate in human repair.

They:

- operate only on the representations provided
- do not infer unstated intent beyond pattern matching
- do not detect conceptual inconsistency unless it is explicitly encoded
- do not refuse to proceed when inputs are ambiguous

Given underspecified or inconsistent inputs, they produce outputs that are:

- locally plausible

- syntactically correct
- semantically ungrounded

Crucially, these outputs are not obviously wrong. They fit the shape of what is expected. They pass superficial inspection. They can even be integrated into downstream processes.

But they carry forward the ambiguity of their inputs, now expressed more formally and at greater scale.

Where a human system would have paused, asked a question, or corrected course, the AI system proceeds.

The result is not immediate failure. It is gradual divergence.

4. The Failure Mode: Local Coherence, Global Incoherence

The characteristic failure mode of AI-assisted workflows in undefined environments is not error, but drift.

Each step in the process produces output that appears reasonable in isolation:

- a requirement that reads clearly
- a design that follows from that requirement
- an implementation that satisfies the design

At no point is there an obvious break.

And yet, across steps:

- the requirement may not reflect the actual stakeholder need
- the design may embed unexamined assumptions
- the implementation may faithfully realise those assumptions

The system is internally consistent and externally wrong.

This failure mode is difficult to detect because it does not manifest as a violation of syntax or logic. It manifests as a mismatch between the system and the domain it is intended to serve.

In domains where precision matters — finance, healthcare, infrastructure — this is not a minor defect. It is a structural risk.

5. A Counterexample: What Must Be True for AI to Work

It is possible to construct workflows in which AI systems participate safely and effectively. But the conditions are stricter than commonly assumed.

In practice, this requires:

- Decomposition of work into explicit transformations: Not “roles” such as “developer” or “analyst,” but functions such as “translate domain intent into testable scenarios.”
- Artefacts as semantic contracts: Documents that encode intent, constraints, and scope in a way that can be consumed without inference.

- Explicit authority boundaries: Clear statements of what the AI may produce and what it may not decide.
- Enforced separation of concerns: Distinct phases (e.g. design, specification, implementation) with defined outputs and prohibitions on mixing them.
- Deliberate context selection: Each interaction with the AI is scoped to the minimum necessary information, reducing signal dilution and unintended coupling.

Under these conditions, AI systems do not replace human judgement. They amplify well-defined transformations.

The difference is not in the model. It is in the structure of the work.

6. The Real Constraint: Work That Cannot Be Stated Cannot Be Externalised

The introduction of AI into a workflow is an act of externalisation. It asks:

“Can this activity be expressed in a form that another system can execute without shared context or tacit knowledge?”

For many organisational activities, the honest answer is no. Not because the activity is inherently inexpressible, but because it has never been expressed. When such activities are handed to AI systems, the result is not automation but amplification of ambiguity. The system produces output. The output is used. The gap between intention and execution widens.

7. Reframing the Problem

The question is not: “How do we adopt AI?”

The question is: “Which parts of our work are sufficiently defined to be externalised, and which are not?”

This reframing has practical consequences:

- Some work will need to be modelled before it can be automated.
- Some work will need to remain human, not by choice but by necessity.
- Some apparent efficiencies will need to be rejected because they bypass required clarification.

AI does not remove the need for conceptual clarity. It makes its absence operationally visible.

8. Conclusion

Most organisations cannot safely use AI — not because the models are insufficient, but because their work is not actually defined.

Human systems tolerate this condition because they rely on continuous, informal repair. AI systems remove that safety net. They operate only on what is made explicit and propagate whatever they are given.

The introduction of AI does not create ambiguity. It reveals it and scales its consequences.

Organisations that wish to use AI effectively must therefore begin not with models, but with their own work:

- What are we actually doing?
- What transformations does it involve?
- What is explicit, and what is assumed?
- Where does authority reside?

Until these questions have answers, AI will not produce clarity. It will produce fluent, ungrounded output that obscures the absence of it. The constraint is not technical. It is architectural. And it has been there all along.